2020A F=ma Exam: Problem 24

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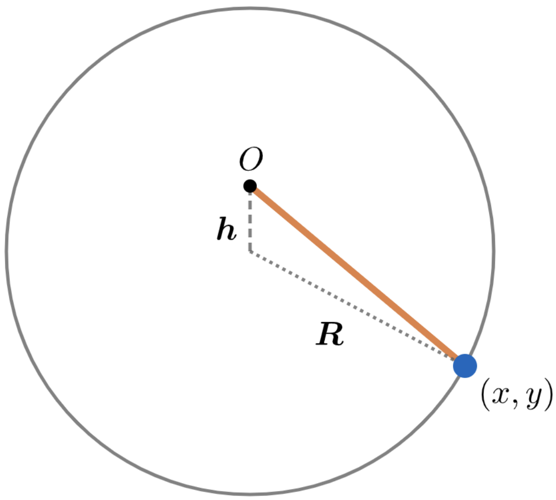

In the absence of gravity, the spring force at (x, y) where the origin O is set at the pole is

$$\vec{F} = (-kx, -ky)$$

and the mass swings in a circle around the origin. With gravity,

$$\vec{F} = (-kx, -ky - mg)$$

Shifting our y-coordinate, $y' = y + mg/k$, we have

$$\vec{F} = (-kx, -ky')$$

and we see now that the mass swings in a circle around the origin in the (x, y') coordinates. Thus,

$$h = \frac{mg}{k}$$

so the answer is $\boxed{\text{D}}$.